# Task 1

## Know your customers

Take-home assessment

Roughly two-thirds of your total time budget goes here.

We run the marketplace pharmacies use to order medicines from many suppliers in one place. Every supplier records the same pharmacy in its own system, under its own name, with its own customer number, and with its own idea of where that pharmacy is. What you will receive is a sanitized extract of exactly that. It is not curated and it is not clean.

## What we want to see at the end

One sentence: **we open your website, click around, and finally see our own customers clearly** — who each pharmacy really is despite ten suppliers calling it ten different names, where it actually is, and how much it buys — with numbers we can trust because your code reproduces them.

## Ground rules

1. **Offline code only.** No network calls, no model APIs inside submitted code. Any offline library is fine.
2. **One command reruns everything.** Every number on your site must be produced by code you hand us, deterministically. We re-run your pipeline and compare it against what the site shows.
3. **Honesty about gaps.** Partial and true beats complete and invented.
4. **Ask, then proceed.** If a definition matters and this document is silent, ask early, state your assumption, keep moving.
5. **Trust nothing blindly** — including the data, and your own first results.
6. Using AI assistants is allowed and expected; keep an honest `AI_DISCLOSURE.md`. Fabricating results or disclosure ends the conversation.

## The data

CSVs for Task 1 sit in `task1/dataset/` in this repository. Column meanings are yours to establish: some are self-evident, some you confirm by testing against the data, and a few only reveal themselves through connections between files. Window: September 2024 to August 2026. Currency: EGP.

| File | Contents |
|---|---|
| pharmacy_registry.csv | our master record of client pharmacies |
| areas_reference.csv | the list of areas we operate in, with map centers |
| supplier_account_names.csv | each supplier branch's own naming book for customer accounts |
| supplier_branches.csv | branches, order minimums, payment terms, discount conventions |
| invoices_app.csv / invoice_details_app.csv | orders placed in our mobile app |
| invoices_erp.csv / invoice_details_erp.csv | orders billed through a partner ERP |
| invoices_legacy.csv / invoice_details_legacy.csv | the oldest system; also where credit documents live |

## The deliverable: a working website on clean data

Structure the files however you like — we do not grade file shapes, we grade what we can *do* with your work. Your `task1/` folder on your fork must contain your cleaning pipeline, and one thing more:

**A `run.bat` in the root of your task1 folder.** We double-click it, a website opens on localhost, and everything below is clickable. If it does not launch on a clean machine, nothing else gets graded. Any web stack is welcome — Streamlit will get you there fastest.

When we browse it, here is what we expect to be able to do, and what "correct" means for each:

1. **One identity per pharmacy.** We click any pharmacy and see every name every supplier uses for it, side by side with the one clean name you settled on. The same pharmacy bought from up to ten suppliers — that is ten different spellings and ten different customer codes for one real shop, and gathering them is the heart of this task. Text that cannot be matched honestly stays unmatched, and the site shows how much of the data that is: a measured unknown beats a forced guess, and we re-run your matching to check.

2. **Every pharmacy in its true area.** A large share of our registry lost its area entirely. The invoices remember it: the delivery addresses name the area — misspelled, abbreviated, buried between street and landmark. Resolve them against the areas list; after your work, browsing the site shows every pharmacy in its true area. We will compare your recovered areas against ours.

3. **The numbers, correct and clickable.** Pharmacies per area. Revenue per area. Top pharmacies overall, and the top pharmacies inside each area. Top suppliers. Date filters and area filters that actually filter. The totals must hold up when we re-run your pipeline — and the three sources disagree about what counts as revenue, so the cleaning rules you chose are yours to make and to document in your README, with the row counts and money impact of each rule.

4. **Drill-down that proves it.** Click one pharmacy: its aliases at every supplier, its area, its revenue over time. This is where you show us the data is real.

**What "good" means.** A site whose every number survives our re-run of your pipeline, built on sources you first made trustworthy, with every ambiguity either measured or asked about. There is no single correct cleaning rule set — there are defensible ones and indefensible ones, and the difference is documentation. We do not care what your intermediate files look like; we care that the site tells the truth.